Forensic Reconstruction and Physical Interpretation of a LAMMPS Granular Shear Dataset

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June 20, 2026

Contents

1 Objective

This document reconstructs the physical meaning, simulation setup, geometry, material properties, outputs, and analysis workflow of a LAMMPS DEM dataset consisting of approximately 20,001 snapshots stored in Dump and Stress folders.

2 Dataset Inventory

Files identified:

- Dump/
- Stress/
- in.shear_FW.txt
- Shear_Boundary_50x40.txt
- log.shear_fixed_Load_1wall.txt

3 Simulation Type

From the input script:

```
dimension 2
units si
atom_style sphere
boundary p m p
```

Therefore:

- Method: Discrete Element Method (DEM)
- Dimension: 2D
- Units: SI
- Boundary conditions: periodic in x, non-periodic in y

4 Domain Geometry

$$x \in [0.0, 0.5] \text{ m} \quad (1)$$
$$y \in [-0.005, 0.805] \text{ m} \quad (2)$$
$$z \in [-0.005, 0.005] \text{ m} \quad (3)$$

Domain dimensions:

$$L_x = 0.500000000000000 \text{ m} \quad (4)$$
$$L_y = 0.810000000000000 \text{ m} \quad (5)$$
$$L_z = 0.010000000000000 \text{ m} \quad (6)$$

5 Particle Types

Type	Count	Meaning
1	2000	Free grains
2	50	Bottom wall
3	50	Top wall

Total particles:

$$N = 2100$$

6 Wall Geometry

Bottom wall:

$$y = 0.19606911047044200958 \text{ m}$$

Top wall:

$$y = 0.59453900743558196762 \text{ m}$$

Gap height:

$$H = 0.59453900743558196762 - 0.19606911047044200958$$

$$H = 0.39846989696513995804 \text{ m}$$

7 Wall Motion

Input:

```
fix bottombnd bottomwall freeze
fix topbnd topwall move linear 0.026 0.0 0.0 units box
```

Therefore:

Wall	Velocity (m/s)	Direction
Bottom	0.000000000000000	Fixed
Top	0.026000000000000	+x

8 Shear Rate

$$\dot{\gamma} = \frac{0.026000000000000}{0.39846989696513995804}$$

$$\dot{\gamma} = 0.065249326754243 \text{ s}^{-1}$$

9 Particle Properties

Density:

$$\rho = 2600 \text{ kg/m}^3$$

Wall diameter:

$$d_{wall} = 0.0100000000000000 \text{ m}$$

Grain diameter range:

$$d_{min} = 0.00800167937467418 \text{ m} \tag{7}$$
$$d_{max} = 0.0119980068899469 \text{ m} \tag{8}$$

Diameter ratio:

$$\frac{d_{max}}{d_{min}} = 1.499464$$

The assembly is polydisperse.

10 Packing Fraction

Total grain area:

$$A_{particles} = 0.158346 \text{ m}^2$$

Cell area:

$$A_{cell} = 0.39846989696513995804 \times 0.5$$

$$A_{cell} = 0.19923494848256997902 \text{ m}^2$$

Packing fraction:

$$\phi = \frac{0.158346}{0.19923494848256997902}$$

$$\phi \approx 0.794770$$

11 Contact Model

```
pair_style gran/hooke/history
10000000
2857142.8
27000
13500
2.0
1
```

Normal stiffness:

$$k_n = 10000000 \text{ N/m}$$

Tangential stiffness:

$$k_t = 2857142.8 \text{ N/m}$$

12 Time Integration

Timestep:

$$\Delta t = 5.77 \times 10^{-7} \text{ s}$$

Run length:

$$N_{steps} = 90000000$$

Physical simulation time:

$$T = 51.93 \text{ s}$$

13 Total Shear Strain

Wall displacement:

$$\Delta x = 0.026 \times 51.93 = 1.35018 \text{ m}$$

Accumulated strain:

$$\gamma = \frac{1.35018}{0.39846989696513995804}$$

$$\gamma = 3.388410670815871$$

14 Output Structure

Snapshots are written every 4500 timesteps.

Total snapshots:

$$N_{snap} = \frac{90000000}{4500} + 1 = 20001$$

14.1 Dump Files

Stored variables:

Column	Meaning
1	id
2	type
3	radius
4	x
5	y
6	z
7	v_x
8	v_y
9	v_z

14.2 Stress Files

Generated by:

```
compute stress/atom NULL pair
```

Stored variables:

$$c_3[1] \to \sigma_{xx} \tag{9}$$
$$c_3[2] \to \sigma_{yy} \tag{10}$$
$$c_3[4] \to \sigma_{xy} \tag{11}$$

15 Relationship Between Dump and Stress

For a given timestep:

```
Dump_Shear.45000000
Stress_Shear.45000000
```

represent the same 2100 particles and are joined by particle ID.
Combined particle dataset:

$$(x, y, v_x, v_y, \sigma_{xx}, \sigma_{yy}, \sigma_{xy})$$

16 Velocity Profile Investigation

A preliminary coarse profile from Dump_Shear.90000000 produced:

Bin	Mean v_x
0	0.000990271
1	0.001139400
2	-0.000068009
3	-0.000850651
4	-0.004082460
5	-0.006584750
6	-0.004282710
7	0.000801035
8	0.004488360
9	0.008256620
10	0.014660500
11	0.018656300
12	0.021567800
13	0.025116000
14	0.027052500
15	0.024614400
16	0.024893700
17	0.025569900
18	0.025269200
19	0.025895900

The upper region approaches the imposed wall velocity of 0.026 m/s.

17 Steady State Evidence

From the log file:

Initial kinetic energy:

$$KE(0) = 0 \tag{12}$$
$$KE(1500) = 0.013646446 \tag{13}$$

Final values:

$$KE(89995500) = 0.00060756053 \tag{14}$$
$$KE(90000000) = 0.00057495114 \tag{15}$$

This indicates a statistically stationary sheared state.

18 Computational Statistics

$$\text{Loop Time} = 21989.8 \text{ s} \tag{16}$$
$$= 6.1083 \text{ h} \tag{17}$$

Average neighbors per atom:

2.14429

Total neighbors:

$$4503$$

Dangerous builds:

$$0$$

19 Physical Interpretation

The simulation represents a dense granular material subjected to simple shear:

$$\text{Dense Packing} \rightarrow \text{Wall Motion} \rightarrow \text{Force Chains} \rightarrow \text{Stress Transmission} \rightarrow \text{Steady Shear Flow}$$

The dataset provides direct access to:

- Positions
- Velocities
- Particle stresses
- Velocity gradients
- Stress fields
- Continuum constitutive behavior

20 Recommended Analysis Workflow

1. Read Dump and Stress files.
2. Merge using particle ID.
3. Keep Type-1 grains.
4. Compute $v_x(y)$.
5. Compute $\sigma_{xx}(y)$.
6. Compute $\sigma_{yy}(y)$.
7. Compute $\sigma_{xy}(y)$.
8. Apply coarse-graining.
9. Build continuum fields.
10. Develop constitutive/TBNN models.

21 Summary

Quantity	Value
Total particles	2100
Grains	2000
Packing fraction	0.794770
Density	2600 kg/m^3
d_{min}	0.00800167937467418 m
d_{max}	0.0119980068899469 m
Wall velocity	0.026000000000000 m/s
Gap height	0.39846989696513995804 m
Shear rate	0.065249326754243 s^{-1}
Physical time	51.93 s
Total strain	3.388410670815871
Snapshots	20001